

Measured on Assistant responses from a chat dataset ($n = 18,777$, sampled from LMSYS-CHAT-1M), the components of persona space explain between 19.4% and 33.6% of the overall activation variance, across the three models. The remaining activation variation likely contains information related to the content and syntax of the response.

One of the models we used, Gemma 2 27B, has both open-weight base and instruct versions available. With the base model, we took the instruct model’s rollouts and ran the remainder of the pipeline; the resulting PCs are nearly identical to that of the instruct model (Appendix B.3). This suggests that these axes of persona differentiation within LLMs are likely already present in base models and inherited from the pre-training corpus.

2.2 Persona space contains interpretable dimensions

We next turn to understanding the semantic meaning of the principal components (PCs). The PCs can be characterized by the role vectors that have a high or low cosine similarity with the PC direction. We arrived at PC interpretations manually by inspecting the similarity of individual roles with the component axis (Appendix B.2).

Some PCs in persona space look remarkably similar across models, in that they project onto the underlying role vectors similarly. We compared how roles were distributed on each model’s PC by measuring the pairwise correlation between the cosine similarities of how role vectors load onto PCs. This gives us the “role composition” of each PC.

Between all pairs of models, the correlation of role loadings on PC1 is >0.92 , indicating remarkably high similarity. PC1 stands out with fantastical characters on one end (*bard*, *ghost*, *leviathan*) and roles more similar to the Assistant persona on the other (*evaluator*, *reviewer*, *consultant*). We hypothesize that this axis measures deviation from the Assistant persona and analyze it in depth in the next section.

The interpretations of the other PCs are somewhat harder to articulate; we encourage referring to Table 1 to understand them better. Speaking loosely, PC2 appears to span a *collective* to *individual* direction in Qwen and Llama, which have a pairwise similarity of 0.89. Gemma is slightly different from the rest (similarity <0.61) and seems to span *informal* or creative to *systematic* roles. PC3 diverges further between models, such that Qwen’s and Llama’s PC3 have a similarity of 0.56, and both are nearly orthogonal to Gemma’s PC3. Qwen and Llama’s axes roughly span intuitive to analytical roles, but Qwen’s has the connotation of *empathetic* to *blunt* while Llama’s has the connotation of

Table 1: The top 5 role vectors with the highest and lowest cosine similarity with each of the top three role PCs for each model. Bolded roles appear in the same list across all three models, while italicized roles appear for two.

	Gemma 2 27B		Qwen 3 32B		Llama 3.3 70B	
	–	+	–	+	–	+
	ROLE-PLAYING ↔ ASSISTANT-LIKE					
Role PC1	<i>bohemian</i>	engineer	<i>bohemian</i>	<i>evaluator</i>	ghost	interpreter
	<i>trickster</i>	<i>analyst</i>	wanderer	<i>examiner</i>	hermit	tutor
	<i>bard</i>	researcher	<i>trickster</i>	<i>reviewer</i>	wraith	<i>evaluator</i>
	prophet	<i>examiner</i>	exile	<i>analyst</i>	leviathan	<i>reviewer</i>
	romantic	forecaster	actor	<i>consultant</i>	<i>bard</i>	<i>consultant</i>
	INFORMAL ↔ SYSTEMATIC			COLLECTIVE ↔ INDIVIDUAL		
Role PC2	chef	<i>synthesizer</i>	<i>swarm</i>	<i>procrastinator</i>	avatar	<i>teenager</i>
	bartender	theorist	<i>hive</i>	amateur	egregore	amateur
	playwright	perfectionist	egregore	<i>teenager</i>	<i>swarm</i>	fool
	amateur	ambassador	crystalline	improviser	<i>hive</i>	comedian
	podcaster	summarizer	purist	influencer	<i>synthesizer</i>	<i>procrastinator</i>
	SOLITARY ↔ RELATIONAL?		EMPATHETIC ↔ BLUNT?		PASSIONATE ↔ ROBOTIC?	
Role PC3	archaeologist	teacher	caregiver	hacker	<i>idealist</i>	<i>robot</i>
	collector	tutor	counselor	spy	survivor	mathematician
	composer	instructor	empath	saboteur	blogger	virus
	philosopher	teenager	<i>idealist</i>	smuggler	evangelist	examiner
	naturalist	assistant	healer	<i>robot</i>	altruist	void

passionate to robotic. Meanwhile, Gemma’s distinct PC3 looks like an axis distinguishing roles with *solitary*, intellectual pursuits from roles with interactive, *relational* duties.

To gain a different lens on the space of available model personas, we reran our entire pipeline with a list of 240 *traits* instead of roles (Appendix C). Similarly, we find that trait space is also a low-dimensional space with a distinctive PC1; one end has traits we expect the Assistant to exhibit (*conscientious, methodical, calm*) while the other has traits we expect to be discouraged (*flippant, mercurial, bitter*). This corroborates the hypothesis that the “Assistant-ness” of a persona is salient to the model.

2.3 Where is the Assistant?

2.3.1 Projecting default activations into persona space

To answer the question of where the Assistant sits within this persona space, we projected the mean activations across response tokens from the model acting in its default Assistant persona.

The default Assistant activation projects onto one extreme of PC1; by contrast, it projects to intermediate values along the other PCs. We quantified this by taking the top ten PCs and calculating the relative position of the default activation’s loading on each PC, within the range of all role projections (with 0 and 1 corresponding to the two role vectors with the minimum and maximum projections along that component). The minimum distance to either extreme of the default Assistant vector’s projection on PC1 was 0.03 (close to the edge), whereas it was between 0.27 and 0.50 on the remaining PCs (Figure 2). This supports our interpretation that this PC roughly measures a “similarity to Assistant” direction.

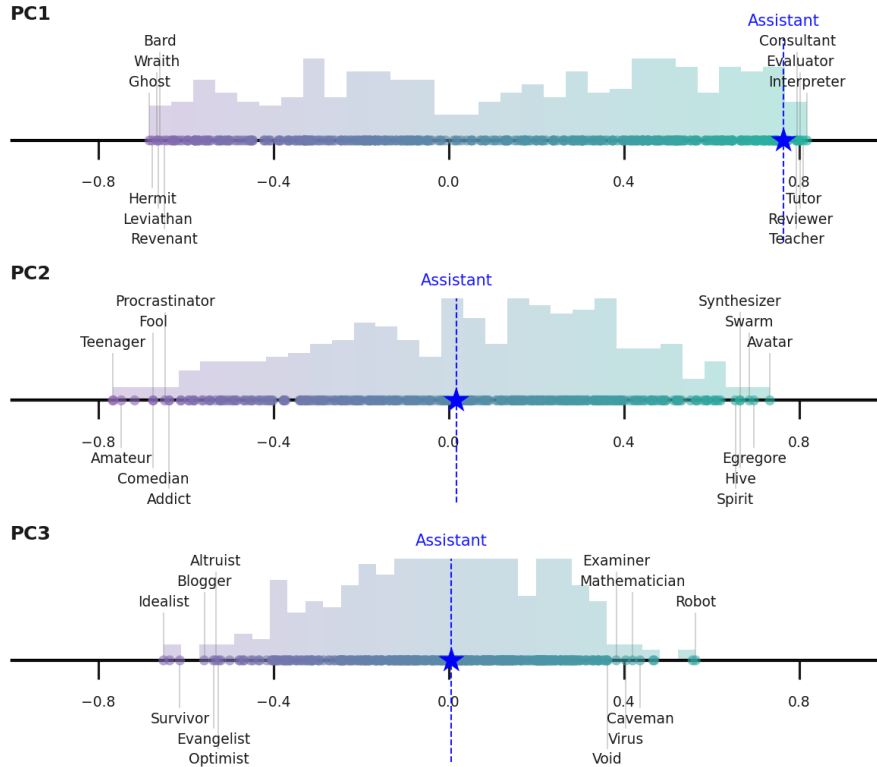


Figure 2: Histogram of cosine similarities of Llama 3.3 70B role vectors with the top 3 PCs from persona space, with selected roles (as well as the default Assistant) labeled.

Table 2: The role and trait vectors with the highest and lowest cosine similarity to the default Assistant activation. Bolded roles are shared across all three models while italicized roles appear for two.

	Gemma 2 27B		Qwen 3 32B		Llama 3.3 70B	
	−	+	−	+	−	+
Roles	<i>fool</i> <i>zealot</i> poet <i>hoarder</i> <i>narcissist</i>	generalist witness specialist <i>synthesizer</i> summarizer	poet leviathan pirate comedian eldritch	<i>interpreter</i> generalist facilitator collaborator moderator	procrastinator <i>fool</i> <i>hoarder</i> <i>narcissist</i> <i>zealot</i>	generalist <i>interpreter</i> <i>synthesizer</i> teacher reviewer
Traits	poetic <i>petty</i> zealous romantic disorganized	calm methodical structuralist moderate conscientious	<i>flippant</i> impulsive nonchalant <i>irreverent</i> goofy	pensive educational meticulous earnest <i>generous</i>	edgy <i>petty</i> <i>irreverent</i> sarcastic <i>flippant</i>	strategic patient diplomatic <i>generous</i> supportive

2.3.2 Cosine similarity with persona vectors

We also measured the cosine similarity between the default Assistant activation and each individual role/trait vector (Table 2). One role that is highly similar to the Assistant persona across models is *generalist*; other common shared roles include *interpreter* and *synthesizer*. Dissimilar roles shared across Gemma and Llama include *fool*, *zealot*, and *narcissist*, while dissimilar roles to Qwen include fantastical or improvisational roles like *poet*, *leviathan*, or *pirate*.

There are some interesting differences across models: Gemma’s Assistant appears emotionally regulated and systematic (*calm*, *methodical*, *structuralist*), Qwen appears pedagogical and thoughtful (*pensive*, *educational*, *meticulous*), and Llama appears socially intelligent and warm (*strategic*, *patient*, *diplomatic*).

3 The Assistant Axis

Inspired by how the main variation in persona space seems to capture how “Assistant-like” the current persona is, we computed a contrast vector between the default Assistant activation and the mean of all role vectors: the Assistant Axis. We can characterize the Assistant Axis by comparing it to our persona vectors and observing the causal effects of steering model activations with this direction.

3.1 Identifying the Assistant Axis

In persona space, PC1 appears to roughly measure how similar roles are to the default Assistant persona. The default response activation projects onto one end of this direction, which is characterized by roles we typically associate with the Assistant (*generalist*, *consultant*, *analyst*). The other end is occupied by human archetypes unlike the Assistant (*hermit*, *pilgrim*, *actor*) and nonhuman entities (*eldritch*, *ghost*, *whale*).

Inspired by these observations, we defined an Assistant Axis as follows. We subtracted the mean of all *fully role-playing* role vectors from the mean default Assistant activation (on the same extraction questions used for the roles) at every layer. We found that the similarity between this vector and PC1 is high: >0.60 at all layers across our three models, and >0.71 at the middle layer of each model (Appendix G.1).

We computed the cosine similarity between the Assistant Axis and our 240 trait vectors in raw activation space and find that traits associated with the Assistant end include *transparent*, *grounded*, and *flexible*, while disassociated traits include *enigmatic*, *subversive*, and *dramatic* (Figure 3).

In the rest of the paper, we perform our experiments with this contrast vector as the Assistant Axis and compare results with using PC1 (Appendix G), finding that the contrast vector is equally as effective for our interventions. We recommend the contrast vector method for reproducing our results in different models because it is not guaranteed that PC1 in every model will correspond to an Assistant Axis.